

LAN 120 Final Project: Sensor Data Investigation

BRIDGE-CI | From Sensors to Scientific Computing

In this final project, you will bring together the major ideas from LAN 120 by designing a small sensor-based investigation. You will build and program an Arduino-based system, collect your own sensor data, analyze and visualize that data in a scaffolded Jupyter notebook, and explain what your findings mean.

Project Purpose

The goal is not to produce a massive dataset or build an advanced machine-learning model. The goal is to demonstrate a sound process from a physical sensor to usable evidence.

Ask a Question → Build → Collect → Prepare → Visualize → Interpret → Connect → Communicate

What You Will Do

1. **Develop a research question:** Choose a question that can be investigated with a sensor and measured over time or across conditions.
2. **Select a sensor:** Choose an Arduino-compatible sensor that can provide data relevant to your question.
3. **Build and test the system:** Create the circuit, verify that the sensor works, and document your setup.
4. **Program the Arduino:** Write or adapt Arduino code to read the sensor and produce usable data. Include meaningful comments in your code.
5. **Plan your data collection:** Decide what you will measure, how often you will sample, how long you will collect data, and what units or labels are needed.
6. **Collect and save the data:** Collect a manageable dataset and save it in a structured format such as CSV.
7. **Analyze the data in Jupyter:** Use the provided scaffolded notebook to load, inspect, summarize, and visualize your data.
8. **Interpret the evidence:** Explain patterns, trends, unusual readings, thresholds, or limitations. Do not simply describe the graph.
9. **Make a cybersecurity or critical infrastructure connection:** Explain how similar sensing and data could be used in a real IoT, industrial, or critical infrastructure context. Consider reliability, data integrity, false readings, or cyber manipulation.
10. **Communicate your findings:** Submit a technical report that clearly explains your question, system, methods, results, limitations, and conclusions.

Project Requirements

- A clear and measurable research question
- An Arduino-compatible sensor and functioning circuit or approved simulation
- Arduino code that reads the sensor and is meaningfully commented
- A documented data-collection plan, including sampling interval or collection method
- A student-generated dataset saved in CSV or another approved structured format
- A completed scaffolded Jupyter notebook using your own data
- At least two meaningful data visualizations, unless otherwise approved
- Interpretation of patterns, trends, anomalies, or other evidence
- Discussion of limitations or possible sources of error
- A cybersecurity, IoT, industrial, or critical infrastructure connection
- A technical report that communicates the investigation and findings

Suggested Sensor Ideas

Temperature, humidity, light, distance, motion, soil moisture, sound, vibration, air quality, pressure, or another instructor-approved sensor. Your project does not have to use a complex sensor. Choose a sensor that helps you investigate a meaningful question.

What Counts as a Good Dataset?

A good dataset is not necessarily a large dataset. It should contain enough observations to help you answer your research question, be organized consistently, and include meaningful labels, units, and time or sample information when appropriate. You should also be able to explain how the data was collected.

Jupyter Notebook Expectations

- Load your dataset successfully.
- Inspect the structure of the data.
- Use the provided code cells and modify the required variables or parameters.
- Create readable, labeled visualizations.
- Calculate or display appropriate summary information when required.
- Explain what the visualizations and results show.
- Identify limitations, unusual values, or possible sources of error.

Technical Report

Your report should include the following sections:

- Research Question and Purpose
- Sensor System and Circuit
- Arduino Program and Data-Collection Method
- Dataset Description
- Jupyter Analysis and Visualizations
- Results and Interpretation
- Limitations and Sources of Error
- Cybersecurity / Critical Infrastructure Connection
- Conclusion and Possible Next Steps

Submission Checklist

- ☐ Final technical report
- ☐ Arduino code
- ☐ Circuit photo, screenshot, or approved demonstration
- ☐ CSV/data file
- ☐ Completed Jupyter notebook
- ☐ Required visualizations

Important Note About AI and Computational Tools

Provided Jupyter notebooks and AI-assisted analysis tools may be used as directed. Their outputs are evidence to examine, not automatic truth. You are responsible for verifying results, explaining your reasoning, documenting required changes to code or parameters, and submitting your own interpretation. Any use of generative AI must follow the instructions for the assignment and be acknowledged when required.

Final Project Rubric — 100 Points

Criteria	Excellent	Proficient	Developing	Incomplete	Pts
Research Question & Project Plan	Clear, measurable question; sensor and collection plan are well matched to the investigation.	Appropriate question and generally clear plan with minor gaps.	Question or plan is vague, overly broad, or only partly measurable.	Research question or meaningful plan is missing.	10
Sensor System & Circuit Design	Sensor is appropriate; circuit functions reliably and is clearly documented and explained.	Circuit functions with minor issues; sensor purpose is adequately explained.	Circuit works inconsistently or documentation/explanation is incomplete.	System is nonfunctional or major components/documentation are missing.	15
Arduino Programming & Data Collection	Code reliably reads the sensor, is meaningfully commented, and supports an appropriate sampling approach.	Code works and collects usable data with adequate comments and sampling.	Code or collection process has errors but produces some usable data.	Code does not produce usable data or required code is missing.	15
Dataset Quality & Documentation	Dataset is organized, usable, labeled, and sufficiently documented; limitations are identified.	Dataset is generally organized and documented with minor omissions.	Dataset requires substantial cleanup or lacks important documentation.	Dataset is missing, unusable, or largely undocumented.	10
Jupyter Notebook Application	Imports own data, runs notebook successfully, makes required modifications, and documents the analysis.	Notebook runs successfully and required modifications are completed.	Notebook is partially completed or errors limit the analysis.	Notebook is missing or does not meaningfully analyze the student's data.	15
Data Visualization & Interpretation	Visualizations are accurate, labeled, and connected to the question; interpretation explains evidence without overstating it.	Visualizations are appropriate and interpretation generally explains findings.	Visualizations or interpretation are incomplete, unclear, or loosely connected to the question.	Meaningful visualizations or interpretation are missing.	15
IoT, Cybersecurity & Critical Infrastructure Connection	Thoughtful connection to real-world sensing, data integrity, cybersecurity, reliability, or operational consequences.	Appropriate connection to IoT, cybersecurity, or critical infrastructure.	Connection is present but superficial or unclear.	Connection is missing or inaccurate.	10
Technical Report & Communication	Clear, professional report covering methods, results, visuals, conclusions, limitations, and next steps.	Report contains required sections and communicates the project adequately.	Report is incomplete, difficult to follow, or missing important sections.	Report is substantially incomplete or missing.	10

TOTAL: 100 points